

Exercise Set 5

Problem 1

A regression estimates the number of road accidents (acc) using daily time series data. The following explanatory variables are included: temperature ($temp$, measured on the Celsius scale) and a dummy variable indicating weekends ($weekend = 1$, if day of the week is Saturday or Sunday; $= 0$ otherwise). Temperature squared is also included in the model ($tempsq$). The model specification is:

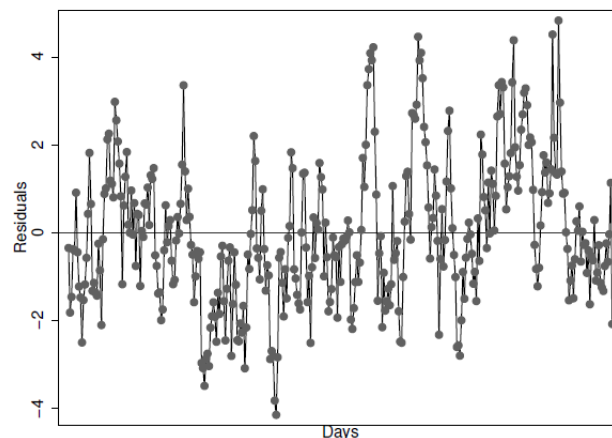
$$\ln(acc_t) = \beta_1 + \beta_2 temp_t + \beta_3 tempsq_t + \beta_4 weekend_t + \varepsilon_t$$

The Stata output below shows results from an OLS estimation.

Source	SS	df	MS	Number of obs = 365		
Model	128.921938	3	42.9739794	F(3, 361) = 15.63		
Residual	992.318274	361	2.74880408	Prob > F = 0.0000		
Total	1121.24021	364	3.08033025	R-squared = 0.1150		
				Adj R-squared = 0.1076		
				Root MSE = 1.658		

ln_acc	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
temp	-.1006878	.0166068	-6.06	0.000	-.133346	-.0680295
tempsq	.0044969	.0007287	6.17	0.000	.0030639	.00593
weekend	-.5620591	.1924058	-2.92	0.004	-.9404361	-.1836821
_cons	7.190896	.1288277	55.82	0.000	6.937549	7.444243

- 1.1 Give the exact interpretation of the estimated coefficient on *weekend*.
- 1.2 Interpret the effects of temperature for a given temperature of 10 degrees Celsius.
- 1.3 The graph below shows the residuals obtained from the regression. Interpret the pattern of the residuals.



- 1.4 Carry out a Breusch-Godfrey test of autocorrelation. Test for a first-order autoregressive process at the 5% significance level. Give the null and alternative hypotheses, test statistic (including the auxiliary regression), degrees of freedom, rejection rule, and your test decision. *Note:* use the Stata output below, where the variable $L1.e$ denotes the lagged residual e_{t-1} .

Source	SS	df	MS	Number of obs = 364		
Model	636.157566	4	159.039391	F(4, 359) = 160.36		
Residual	356.042705	359	.991762409	Prob > F = 0.0000		
Total	992.200271	363	2.73333408	R-squared = 0.6412		
				Adj R-squared = 0.6372		
				Root MSE = .99587		

e	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
L1.e	.8033894	.0317211	25.33	0.000	.741007	.8657719
temp	.0034219	.0099779	0.34	0.732	-.0162006	.0230443
tempsq	-.0000395	.0004385	-0.09	0.928	-.0009019	.0008228
weekend	.1284414	.1157486	1.11	0.268	-.0991891	.3560719
_cons	-.0647561	.0774248	-0.84	0.404	-.2170193	.0875071

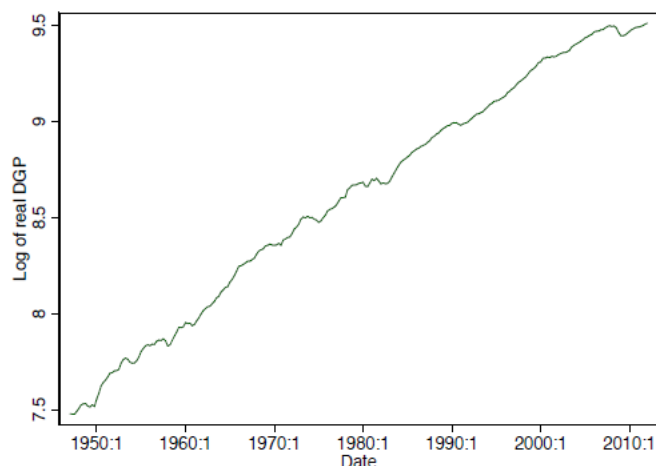
Problem 2

The Phillips Curve describes the inverse relationship between inflation and unemployment. Use the dataset `phillips` in the R-package `wooldridge` to analyze whether policy makers face a trade-off between these two economic variables.

- Make yourself familiar with the dataset. Plot the unemployment rate and the inflation rate over time.
- Test whether there is a trade-off between unemployment and inflation. Use (i) data up to 1996, (ii) 1997-2003 and (iii) the whole time series. Report your results in a table.
- Plot the residuals obtained from the regression and interpret the pattern.
- Test for first-order autocorrelation. Carry out a t-test, a Breusch-Godfrey Lagrange multiplier test and a Durbin-Watson test.
- Use the entire data set. Estimate the Phillips curve model by iterative Cochrane-Orcutt and Prais-Winsten. How similar are the final estimates of ρ ? How similar are the PW and CO estimates of β_1 ? Compare ρ and β_1 also to the respective OLS estimation results.
- Include inflationary expectations into your model. Assume adaptive expectations, i.e. the expected value of current inflation depends on recently observed inflation. Estimate the following model: $\Delta inf_t = \beta_0 + \beta_1 unemp_t + e_t$. Test for first-order autocorrelation.
- Estimate a Phillips curve model where inflation and unemployment are in first differences.

Problem 3

- 3.1 The graph below shows the log of the United States real GDP over the period 1947 to 2012. Is the series stationary? Explain.



- 3.2 Show $E(\varepsilon_t)$ for an AR(1) process. Make the following assumptions: (i) $\varepsilon_t = \rho\varepsilon_{t-1} + v_t$, (ii) $Var(v_t) = \sigma_v^2$, and (iii) Gauss-Markov assumptions are satisfied for v_t .

Problem 4

- 4.1 Explain the autocorrelation function (ACF) and the partial autocorrelation function (PACF) of an autoregressive process of order p .
- 4.2 Explain the autocorrelation function (ACF) and the partial autocorrelation function (PACF) of a moving average process of order q .
- 4.3 Suppose you want to investigate the time series properties of the inflation rate in the US over the years from 1960 to 2010 by means of an augmented Dickey-Fuller (ADF) test. (i) State the null hypothesis and the alternative hypothesis. (ii) Specify the ADF test regression when two lags are included. (iii) Carry out the ADF test using the Stata output below ($\alpha = 0.1$). (iv) Give your decision.

Augmented Dickey-Fuller test for unit root			Number of obs =	201
			----- Interpolated Dickey-Fuller -----	
	Test	1% Critical	5% Critical	10% Critical
	Statistic	Value	Value	Value
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Z(t)	-3.145	-3.476	-2.883	-2.573
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4.4 Compute a measure of inflation persistence using the estimation results from the AR(3) model shown below. What is your conclusion?

ARIMA regression

Sample: 1960q2 - 2011q1

Log likelihood = -463.6308

Number of obs = 204

Wald chi2(3) = 283.64

Prob > chi2 = 0.0000

	infl	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	

infl							
	_cons	3.781947	1.020659	3.71	0.000	1.781492	5.782402

ARMA							
	ar						
	L1.	.2925791	.0579182	5.05	0.000	.1790614	.4060968
	L2.	.2277573	.0613714	3.71	0.000	.1074715	.3480431
	L3.	.2969597	.0534582	5.55	0.000	.1921836	.4017358

	/sigma	2.342933	.0538576	43.50	0.000	2.237374	2.448492

4.5 Suppose a Ljung-Box test for 12 lags is carried out for an AR(3) model. The empirical value of the test statistic is 16.67. Give the null hypothesis and the alternative hypothesis, number of degrees of freedom, rejection rule ($\alpha = 0.01$), and test decision.

Problem 5

The data set INCOME.dta contains information on quarterly disposable income in the United Kingdom for the quarters 1971:I to 1985:II, measured in million pounds and current prices ($T = 58$).

- 5.1 Produce a graph of the natural logarithm of quarterly income.
- 5.2 Perform augmented Dickey-Fuller tests including 1 up to 6 lags, with and without including a linear trend. What do you conclude about the presence of a unit root in log income?
- 5.3 Transform the series into first differences and produce a graph. Perform augmented Dickey-Fuller tests on the change in log income including 1 up to 6 lags. What do you conclude? Motivate why you did or did not include a time trend.
- 5.4 Determine the sample ACF and PACF for the change in log income. Is there an obvious model suggested by these graphs?
- 5.5 Estimate an ARIMA(4,1,4) model for the change in log income.
- 5.6 Test for serial correlation in the residuals of the model. Can you reject the null hypothesis of white noise errors?

References

- Wooldridge, J. (2013). *Introductory Econometrics. A modern approach*.
- Verbeek, M. (2012). *A Guide to Modern Econometrics*. John Wiley & Sons, Chichester.